

Project Proposal: Pedestrian Action & Intent Recognition on TITAN

AER 1515: Perception for Robotics

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I. PROBLEM OUTLINE & APPLICATIONS

Objective. Build a *practical* perception pipeline that, from ego-view video, detects pedestrians and classifies their *near-term intent* into three classes: *waiting-to-cross*, *crossing*, and *walking-along* (not intending to cross). We explicitly *do not* tackle long-horizon forecasting; we target short tracklets (approximately 1–2 s) and frame-to-few-second intent.

Why it matters. Short-horizon pedestrian intent improves proactive braking/yielding in urban driving and can transfer to mobile robots operating near people (delivery, assistive, or social robots).

Achievability. We restrict labels to a small, well-populated subset, use a light detector, a simple tracker, and a compact temporal head. We report clear metrics and qualitative overlays.

Assumptions. RGB video only; IMU/odometry may be used only for ablation if time permits. No SLAM; no map priors.

II. RELATIONSHIP TO COURSE CONTENT

Our design implements course concepts across modules:

Module 1 (Foundations).

- *Probability/Information/Optimization*: cross-entropy loss, regularization, and early stopping; confidence calibration for intent outputs.
- *ML in a Nutshell*: supervised learning with train/val/test split and data augmentation; reproducible experiments.
- *Sensor Modelling*: we reason about ego-camera characteristics (frame rate, FOV, motion blur) and their impact on detection/temporal features.
- *Computer Vision*: detection backbones and spatio-temporal features from image patches.

Module 2 (Static World).

- *Image Features/Matching*: we use learned features; baseline sanity checks with classical keypoints for motion cue intuition (if time).

Module 3 (Dynamic World).

- *2D Classification/Detection/Segmentation*: pedestrian detection fine-tuning and per-frame *action/intent* classification.
- *3D Tracking/Prediction*: we implement simple 2D tracking (ByteTrack) to create short-term trajectories (a precursor to prediction) and evaluate intent on tracklets.

III. SOLUTION METHOD

Task Definition. Given a video clip sampled at 10 Hz, detect pedestrians each frame, associate them into tracklets of 10–20 frames, and classify each tracklet into one of three intents: {waiting-to-cross, crossing, walking-along}.

A. Data Ingestion & Label Mapping

- **Sampling**: decode at 10 Hz (provided by annotations); resize frames to 1280×720 for speed.
- **Detections**: train/fine-tune a lightweight detector (YOLOv8n) on TITAN person boxes (train split only).
- **Tracklets**: associate detections using ByteTrack; keep tracklets 10–20 frames long (1–2 s).
- **Intent labels (3-class)**. Map TITAN pedestrian action sets to: *waiting-to-cross* $\leftarrow$ labels containing “waiting to cross”; *crossing* $\leftarrow$ labels containing “crossing”/“jay-walking”; *walking-along* $\leftarrow$ atomic/simple-contextual walk on sidewalk/side-of-road without cross intent. Ambiguous/rare classes $\rightarrow$ exclude or map to nearest of three after a quick frequency check.

B. Models

- **Detector (Input Quality)**. YOLOv8n, batch size 16, cosine LR, 50 epochs or early stop by val mAP. Augmentations: mosaic off (stability), HSV jitter, flip.
- **Baseline (Frame-level)**. 2D CNN feature (e.g., MobileNetV3-small) on person crops $\rightarrow$ MLP classifier for 3-way intent. Majority-vote over a tracklet.
- **Temporal Model (Primary)**. 3D ResNet-18 (clip length 16 frames @ 10 Hz; crop size 112×112) or 2D CNN + GRU (hidden 128, 1 layer). Loss: weighted cross-entropy (handle class imbalance). Regularization: dropout 0.3; label smoothing 0.05.

C. Training & Evaluation

- **Split**: use canonical 400/200/100 clips (train/val/test).
- **Metrics**: macro-precision/recall/F1 (primary), per-class F1, and a 3×3 confusion matrix. Report detector mAP@0.5 on val to confirm input quality.
- **Qualitative**: overlay predicted intent on bounding boxes for selected clips (for initial results + final video).

D. Compute & Code Hygiene

Run on a single desktop GPU (or Colab Pro if needed). Fix random seeds, save configs (YAML), and log runs (learning

curves, confusion matrices). Export a small demo script for the showcase video.

IV. DATASET: HONDA TITAN [1]

Collection. Ego-view driving clips recorded in central Tokyo; videos captured with an action camera at high frame rate; synchronized IMU available. Curators provide 700 short clips with annotations at 10 Hz.

Annotations. Frame-wise bounding boxes and tracking IDs for road agents; rich pedestrian action taxonomy (atomic and contextual actions) enabling mapping to crossing-related intent.

Our subset. We use person tracks only and restrict labels to three intent classes defined above to ensure sufficient support and simple evaluation. We follow the official clip split and report results on the held-out test set exactly once.

Preprocessing. Person crops resized to 112×112 for temporal model (and 224×224 for an optional ablation). Tracklets padded/truncated to fixed length (16 frames). Optional: down-sample IMU yaw rate to 10 Hz for ablation (not primary).

V. GOALS & WEEK-BY-WEEK TIMELINE

Two people, 5 hrs/week each $\Rightarrow$ 10 hrs/week total. Each week ends with a concrete artifact (model, plot, or script).

Week	Milestone / Deliverable
Oct 10–16	Setup & Scoping. Verify data access; write label-mapping script for 3-class intent; prepare clip list splits; implement frame decode at 10 Hz.
Oct 17–23	Detector. Fine-tune YOLOv8n on TITAN persons (train); validate on val set; export detector weights; <i>Deliverable:</i> mAP@0.5 curve + sample detections.
Oct 24–30	Tracking. Integrate ByteTrack; generate tracklets (10–20 frames); sanity-check ID persistence; <i>Deliverable:</i> tracklet stats (length histogram) + short overlay video.
Oct 31–Nov 6	Baseline Classifier. MobileNetV3+MLP frame classifier with majority-vote over tracklets; tune class weights; <i>Deliverable:</i> first confusion matrix (val) + qualitative overlay.
Nov 7–13	Temporal Model. Train 3D ResNet-18 (or 2D+GRU) on crops; compare to baseline; <i>Deliverable:</i> macro-F1 (val) and ablation table.
Nov 14	Initial Results Due. Submit 4-page update with methods, metrics, and overlays from the best model so far.
Nov 15–21	Refinement. Error analysis (per-class), simple augmentations (temporal jitter), threshold tuning; start assembling demo clips; <i>Deliverable:</i> improved confusion matrix + better overlays.
Nov 22–27	Demo Video. Script & edit a 3–5 min video with problem, method, and results; finalize visuals.
Nov 28	Demo Video Due. Submit final cut.
Nov 29–Dec 5	Ablations & Write-up. Optional IMU yaw-rate ablation; freeze final model; generate final plots/tables; draft IEEE report.
Dec 6–Dec 12	Polish & Submit. Proof, figure captions, references; archive code/configs; submit 6-page IEEE report.

VI. EXPECTED OUTCOMES & RISKS

Outcomes. (i) A working detector+tracker+temporal intent classifier; (ii) macro-F1 and per-class F1 on held-out test; (iii) annotated video demo; (iv) reproducible code/configs.

Risks & Mitigations. *Class imbalance:* use weighted loss and sampling. *Noisy tracks:* tighten association thresholds and minimum track length. *Time limits:* keep IMU fusion as an ablation; prioritize 3-class mapping and temporal model.

REFERENCES

- [1] S. Malla, B. Dariush, and C. Choi, “Titan: Future forecast using action priors,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 11 186–11 196.